

Database Management Systems

Lecture 1

Database

A database is an organized collection of interrelated data, generally stored and accessed electronically from a computer system.

Database Management System

A DBMS is a software that interacts with end users, applications, and the database itself to control the storage, organization, and retrieval of data.

Database System:

Combining the two (DBMS + Database) makes it a **Database System**.

Communication

To communicate with the Database where the data resides, we will use **Structured Query Language (SQL)**

Key Features of DBMS

- Minimizes data redundancy (repetition)

Key Features of DBMS

- Ensures data consistency

Key Features of DBMS

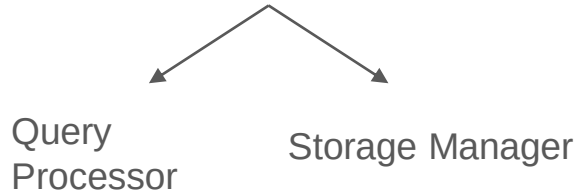
- Easy to write, update, search, delete using SQL
- Provides Atomicity

Key Features of DBMS

- Data Isolation - Data is stored using different tables.
- Security - Data is accessed with username & password
- We can set **integrity constraints**

Different Parts of the Database System

Database System = DBMS + DB



Query Processor

1. DDL Interpreter
2. DML Compiler
3. Query Evaluation Engine

1. DDL Interpreter

Data Definition Language.

Defines the database and the tables, and also used to modify the table.

2. DML Compiler

Data Manipulation Language.

Insert/Update/Delete/Search data from tables

DML Compiler converts SQL statements to low level instructions.

3. Query Evaluation Engine

Executes the low level instructions generated by the DML Compiler

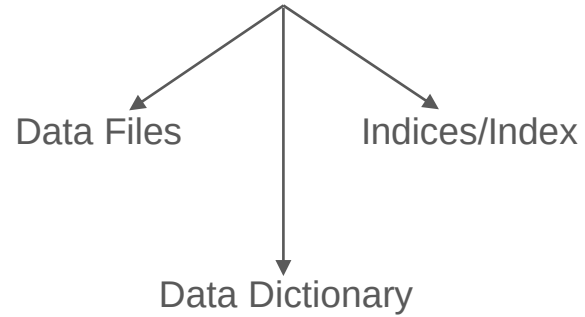
Storage Manager

File Manager. Manages the allocation of space, where to store new data.

Buffer Manager. Most frequently used data in buffer.

Different Parts of the Database System

Database System = DBMS + DB



Data Files

The database itself.

Data Dictionary

Characteristics of the data information are stored.

Indices/Index

For speeding up searches.

Example of a simple Database

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2

A database that stores student and course information.